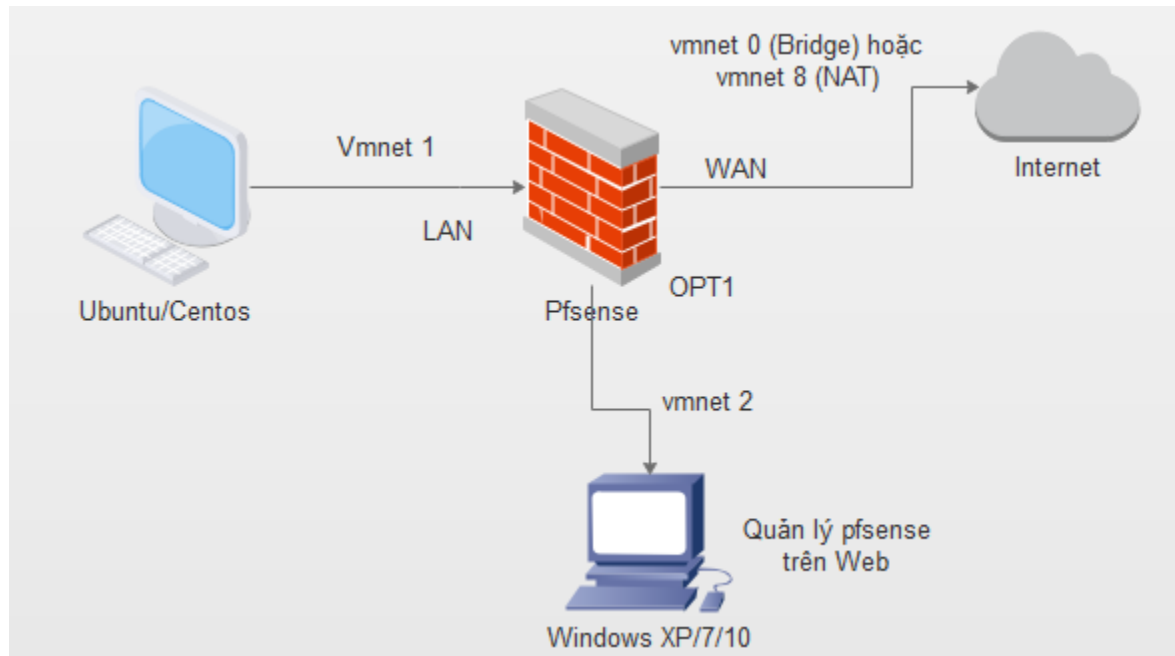


Mô hình lab



Yêu cầu:

1. Cấu hình dịch vụ DHCP trên PfSense, cấp ip động cho máy Ubuntu
2. Cấu hình tính năng NAT trên pfsense, cho phép máy Ubuntu truy cập internet
3. Cấu hình dịch vụ Squid Proxy Server trên firewall pfsense cấm máy Ubuntu truy cập Youtube.com hoặc Facebook.com, các website khác vẫn truy cập bình thường

Hướng dẫn:

1. Cài đặt pfsense và cho phép máy Ubuntu truy cập internet: <https://securityzone.vn/t/lab-pfsense-cau-hinh-nat-tren-pfsense.957/>
2. Cấu hình tính năng Squid Proxy Server trên PfSense, cấm máy Ubuntu truy cập Youtube.com hoặc Facebook.com, Website khác vẫn sử dụng bình thường: <https://github.com/hoangdh/ghichep-pfsense/blob/master/docs/Cau-hinh-chan-website-squidguard.md>

-----Hết-----

Sinh viên thực tập-Viết thành file báo cáo minh họa các bước làm và kết quả đạt được.

Hướng dẫn các bước cài đặt như mô hình đưa ra.

Tải file iso firewall pfsense live cd i386.iso trên trang pfsense

1. Các bước thực hiện:
 - 3.1 cài đặt và cấu hình máy pfsense
 - a. add card mạng



Card vmnet 0: kết nối ra internet

Card vmnet 1, 2: để ở chế độ host only, đồng thời tắt dhcp

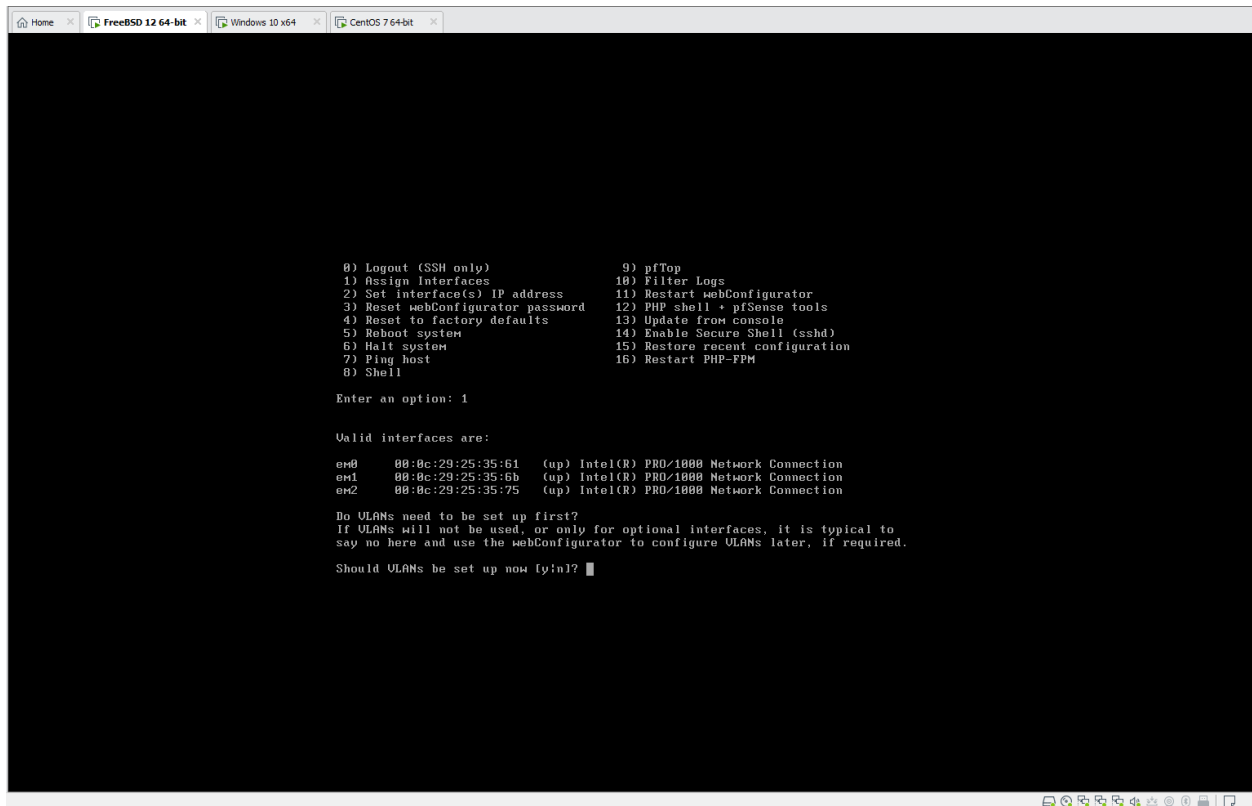
Virtual Network Editor					
Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Qualcomm Atheros AR956x ...	-	-	-
VMnet1	Host-only	-	Connected	-	192.168.118.0
VMnet2	Host-only	-	Connected	-	100.0.0.0

b. gán địa chỉ ip:

interface em0: là card wan

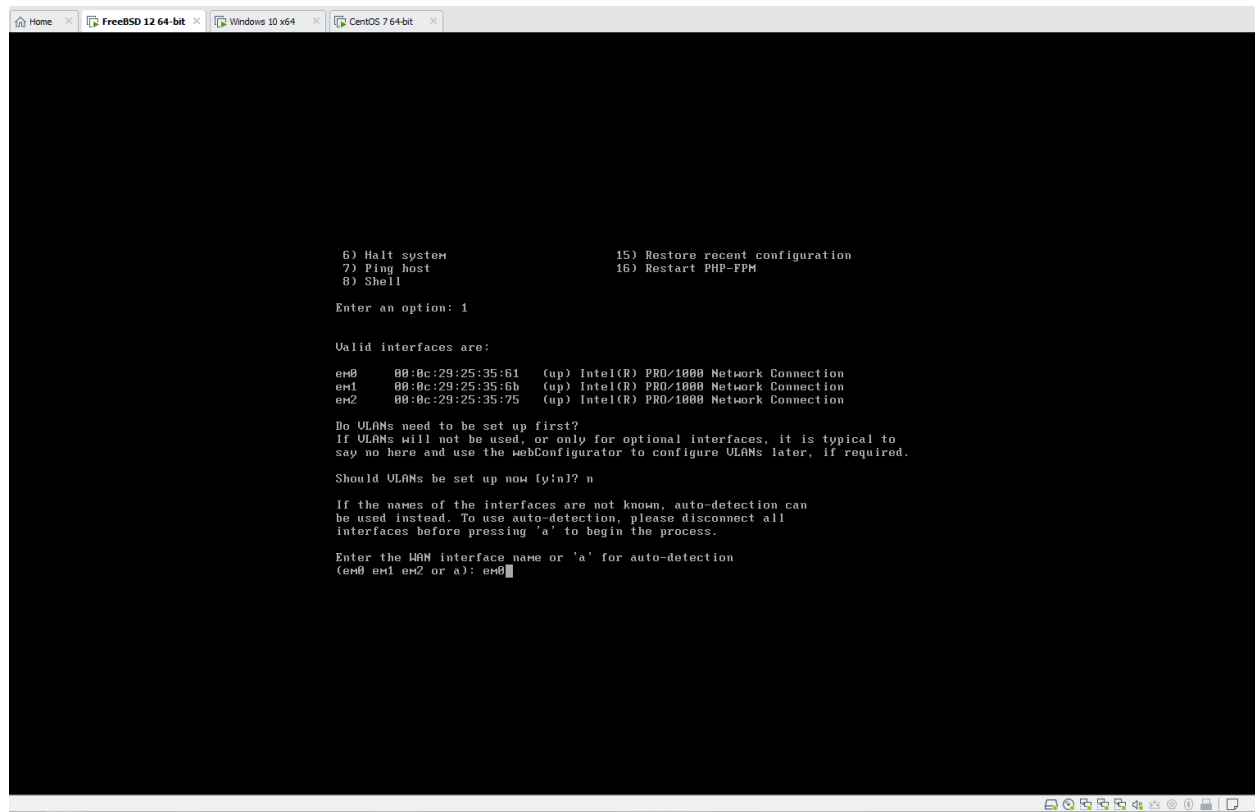
em1: card lan

em2: card opt1



Do you want to set up Vlan's now?

Ta chọn: no



Card wan: em0

```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

Enter an option: 1

Valid interfaces are:

em0      00:0c:29:25:35:61  (up) Intel(R) PRO/1000 Network Connection
em1      00:0c:29:25:35:6b  (up) Intel(R) PRO/1000 Network Connection
em2      00:0c:29:25:35:75  (up) Intel(R) PRO/1000 Network Connection

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y:n]? n

If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 a or nothing if finished): em1
```

Card lan: em1

```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

em0      00:0c:29:25:35:61  (up) Intel(R) PRO/1000 Network Connection
em1      00:0c:29:25:35:6b  (up) Intel(R) PRO/1000 Network Connection
em2      00:0c:29:25:35:75  (up) Intel(R) PRO/1000 Network Connection

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y:n]? n

If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 a or nothing if finished): em1

Optional interface 1 description found: OPT1
Enter the Optional 1 interface name or 'a' for auto-detection
(em2 a or nothing if finished): em2
```

Card opt1: em2

```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

Should VLANs be set up now [y:n]? n

If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 a or nothing if finished): em1

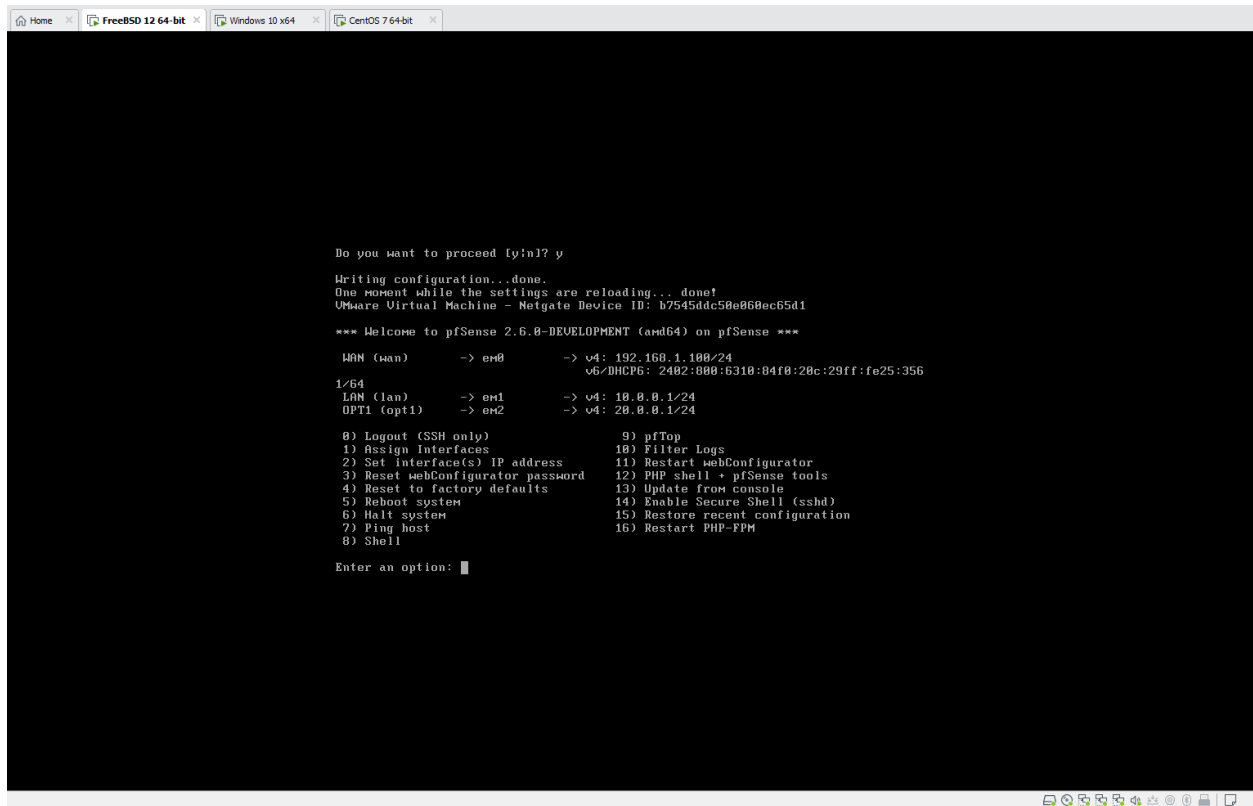
Optional interface 1 description found: OPT1
Enter the Optional 1 interface name or 'a' for auto-detection
(em2 a or nothing if finished): em2

The interfaces will be assigned as follows:

WAN -> em0
LAN -> em1
OPT1 -> em2

Do you want to proceed [y:n]? y
```

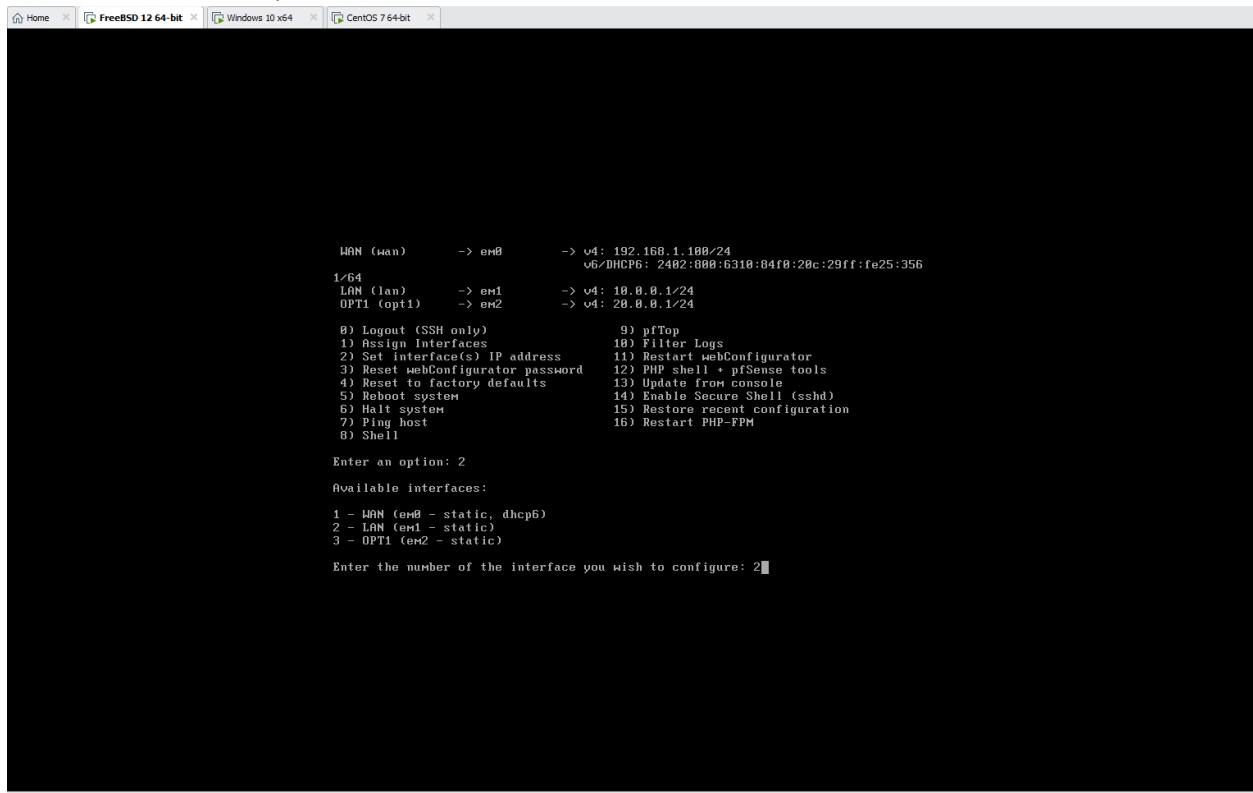
chọn yes: để tiếp tục



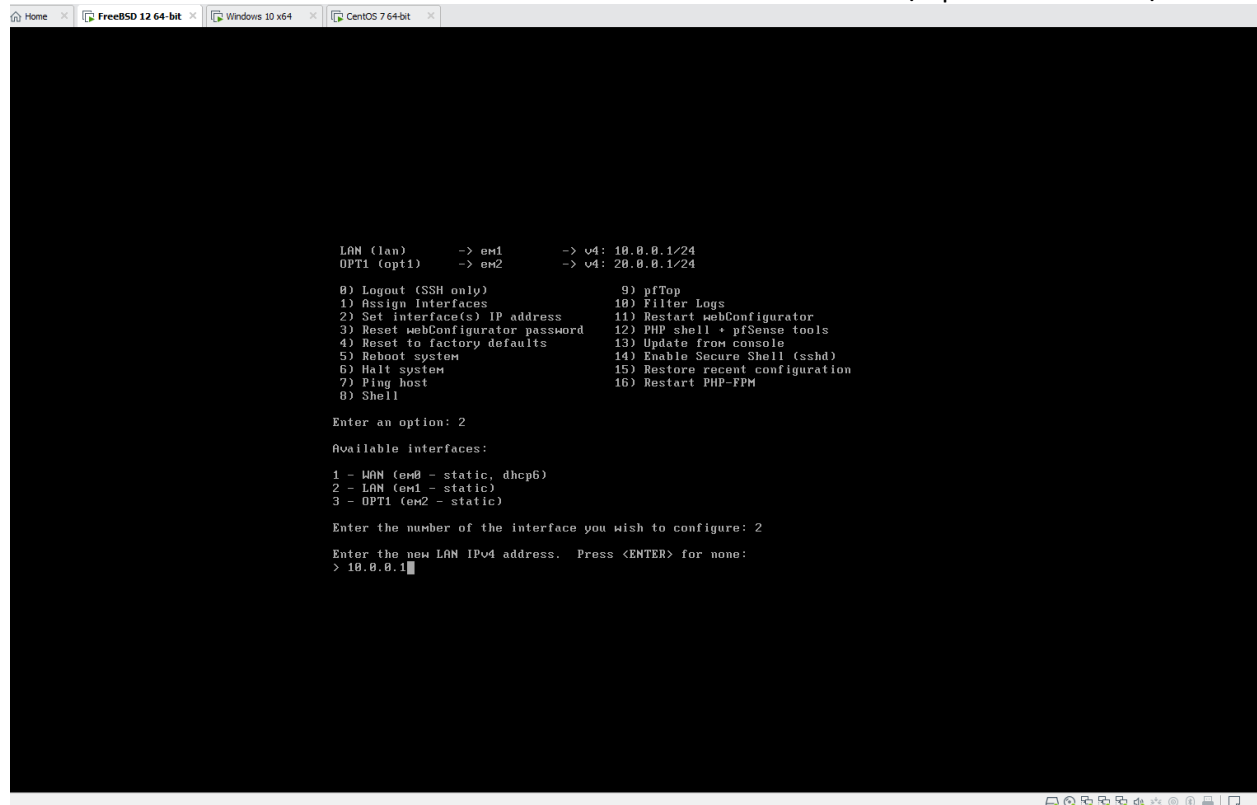
tiến hành đặt ip



Ta chọn số 2: để đặt ip



đặt ip cho card lan: chọn số 2



ip cho card lan là 10.0.0.1/24


```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

1 - WAN (em0 - static, dhcp6)
2 - LAN (em1 - static)
3 - OPT1 (em2 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.0.0.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
```

cấu hình dhcp cho mạng lan: range là: 10.0.0.100-10.0.0.200

```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

3 - OPT1 (em2 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.0.0.1

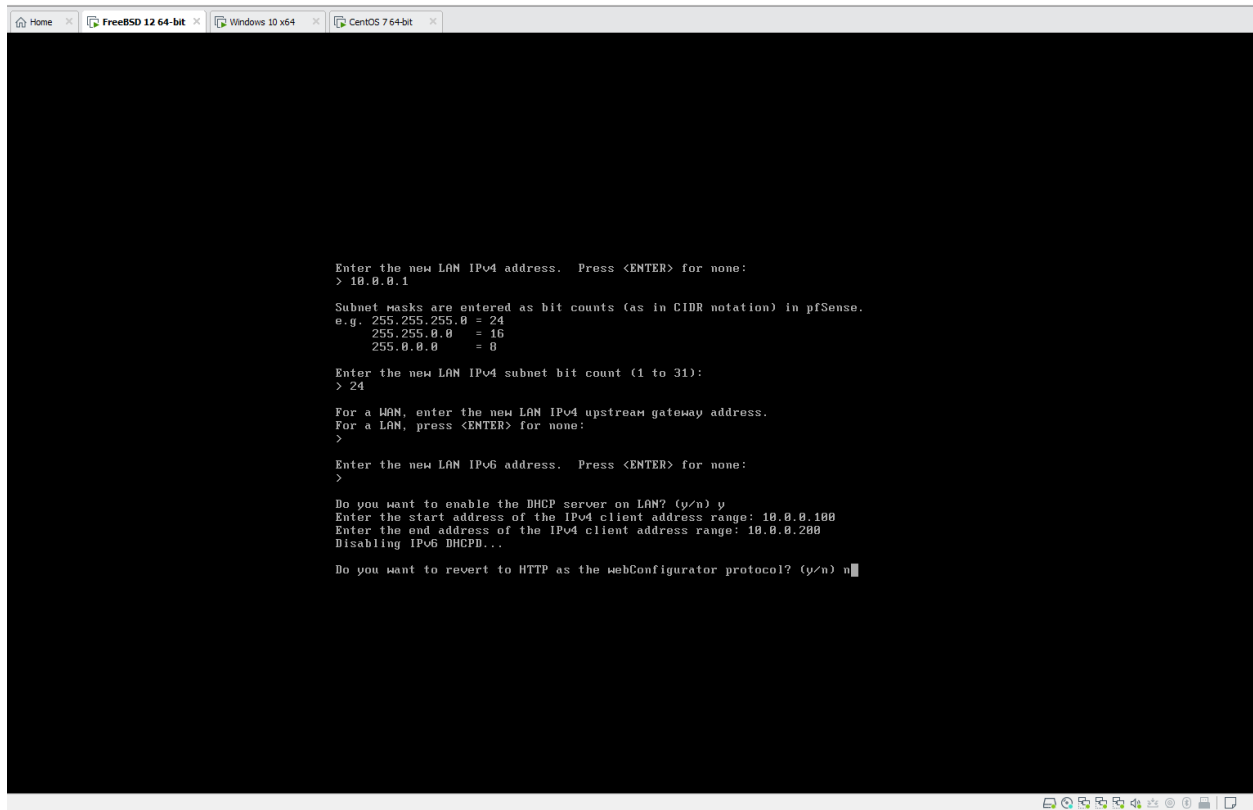
Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

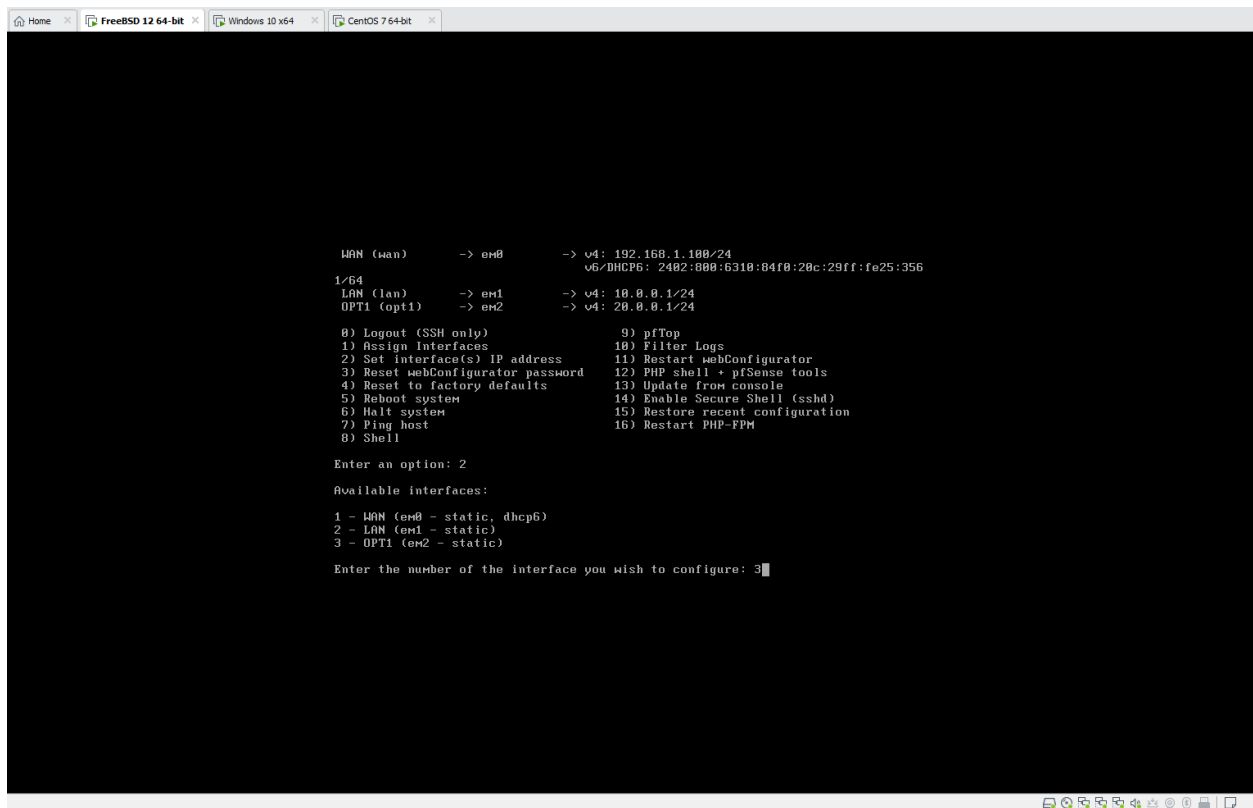
For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 10.0.0.100
Enter the end address of the IPv4 client address range: 10.0.0.200
```



cấu hình card opt1, ta làm tương tự



```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

5) Reboot system          14) Enable Secure Shell (sshd)
6) Halt system            15) Restore recent configuration
7) Ping host              16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - static, dhcp6)
2 - LAN (em1 - static)
3 - OPT1 (em2 - static)

Enter the number of the interface you wish to configure: 3

Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 20.0.0.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 31):
> 24
```

Ip cho card opt1 là 20.0.0.1/24

```
Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

3 - OPT1 (em2 - static)

Enter the number of the interface you wish to configure: 3

Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 20.0.0.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new OPT1 IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on OPT1? (y/n) y
Enter the start address of the IPv4 client address range: 20.0.0.100
Enter the end address of the IPv4 client address range: 20.0.0.200
```

Ip cấp là 20.0.0.100-20.0.0.200

```
8) Shell
Enter an option: 99

VMware Virtual Machine - Netgate Device ID: b7545ddc50e860ec65d1

*** Welcome to pfSense 2.6.0-DEVELOPMENT (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4: 192.168.1.100/24
                v6/DHCP6: 2402:800:6310:84f0:20c:29ff:fe25:356
1/64
LAN (lan)      -> em1      -> v4: 10.0.0.1/24
OPT1 (opt1)    -> em2      -> v4: 20.0.0.1/24

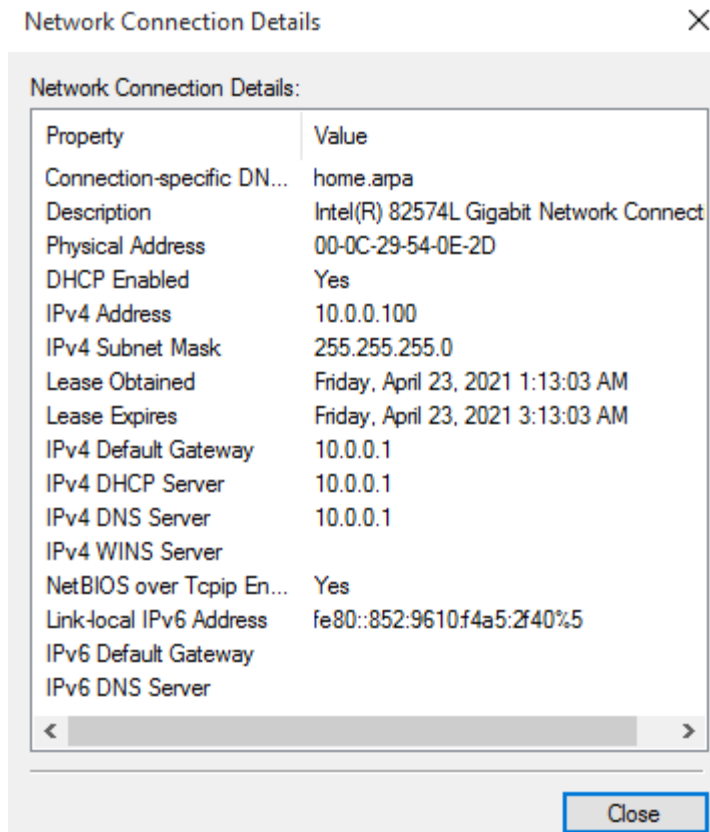
0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: █
```

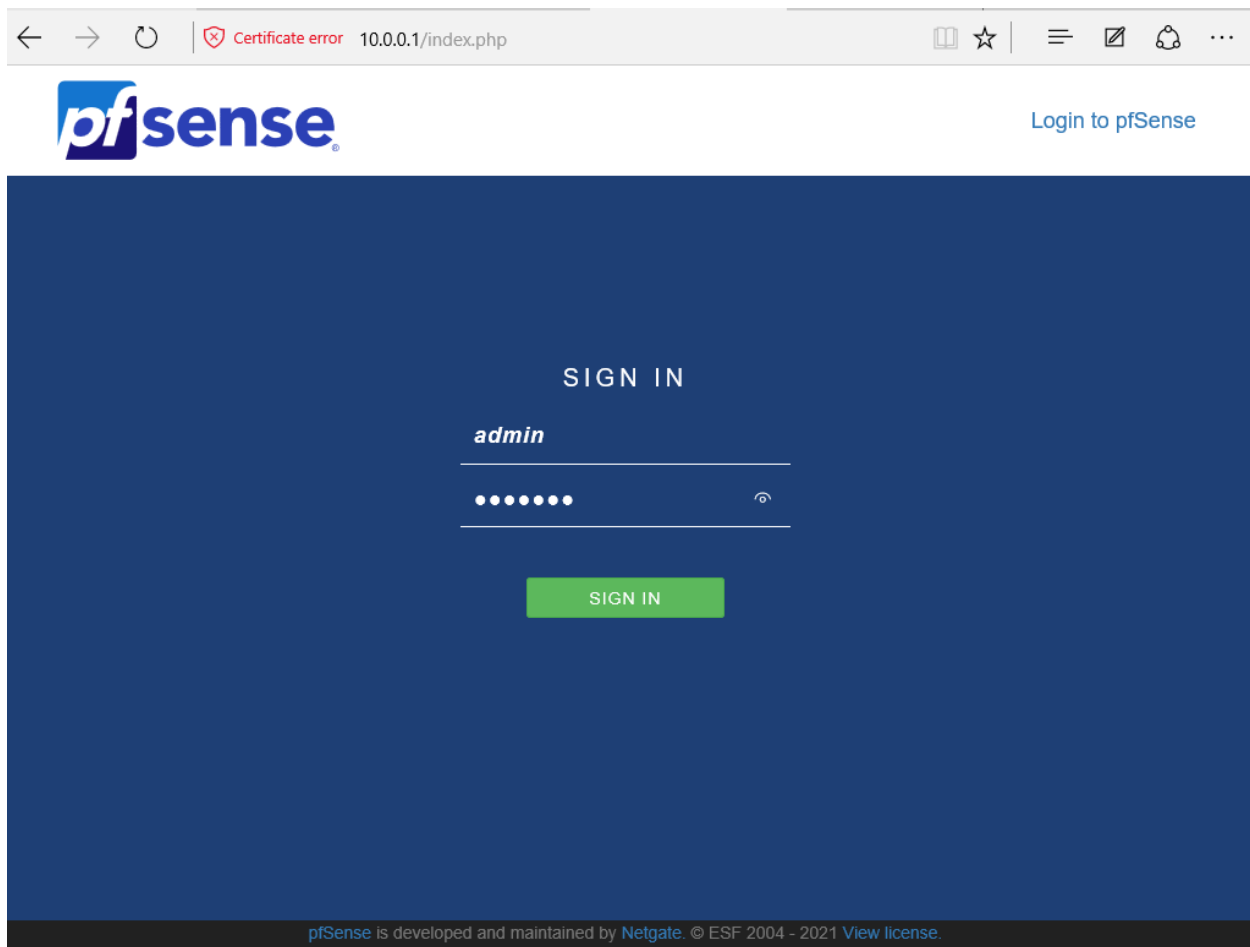
kết quả sau khi đặt ip xong nhấn

99 để lưu cấu hình.

c. cấu hình openvpn trên giao diện web thực hiện
trên windows 10

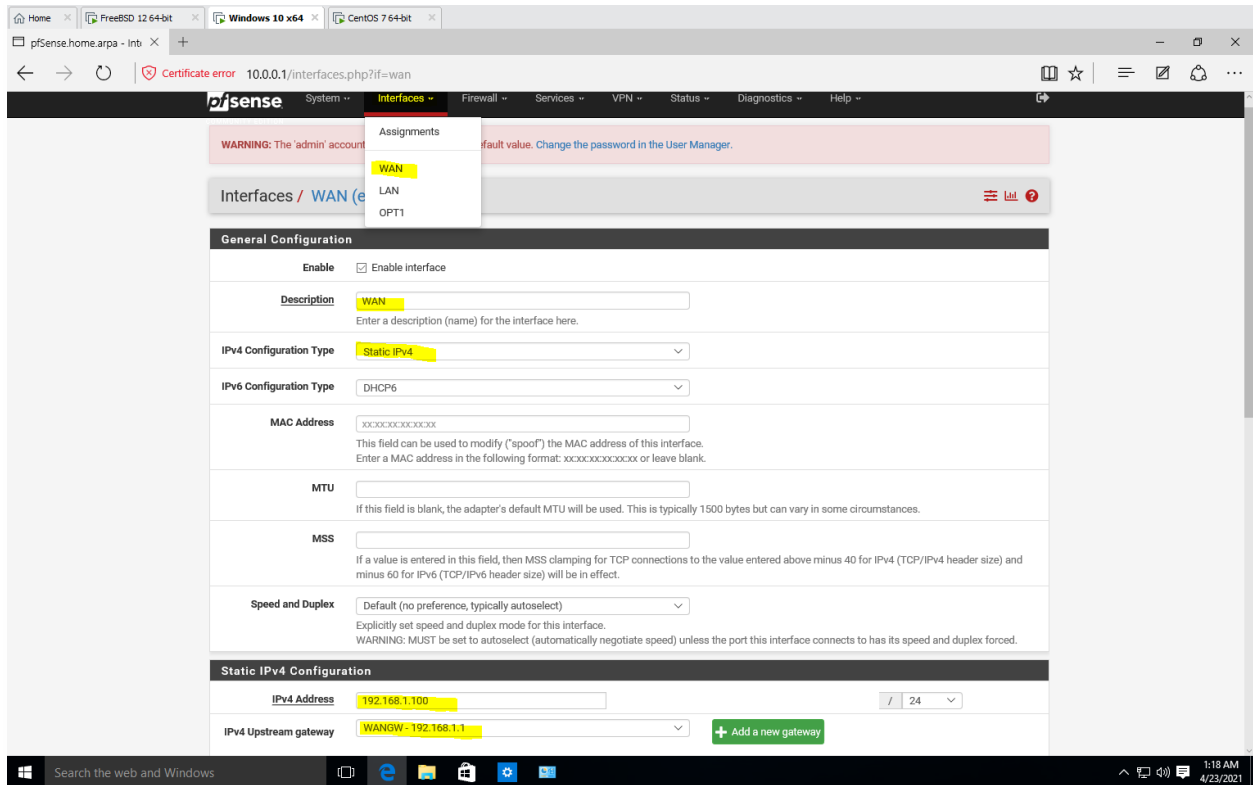


Card vmnet1 nhận ip thành công từ pfsense

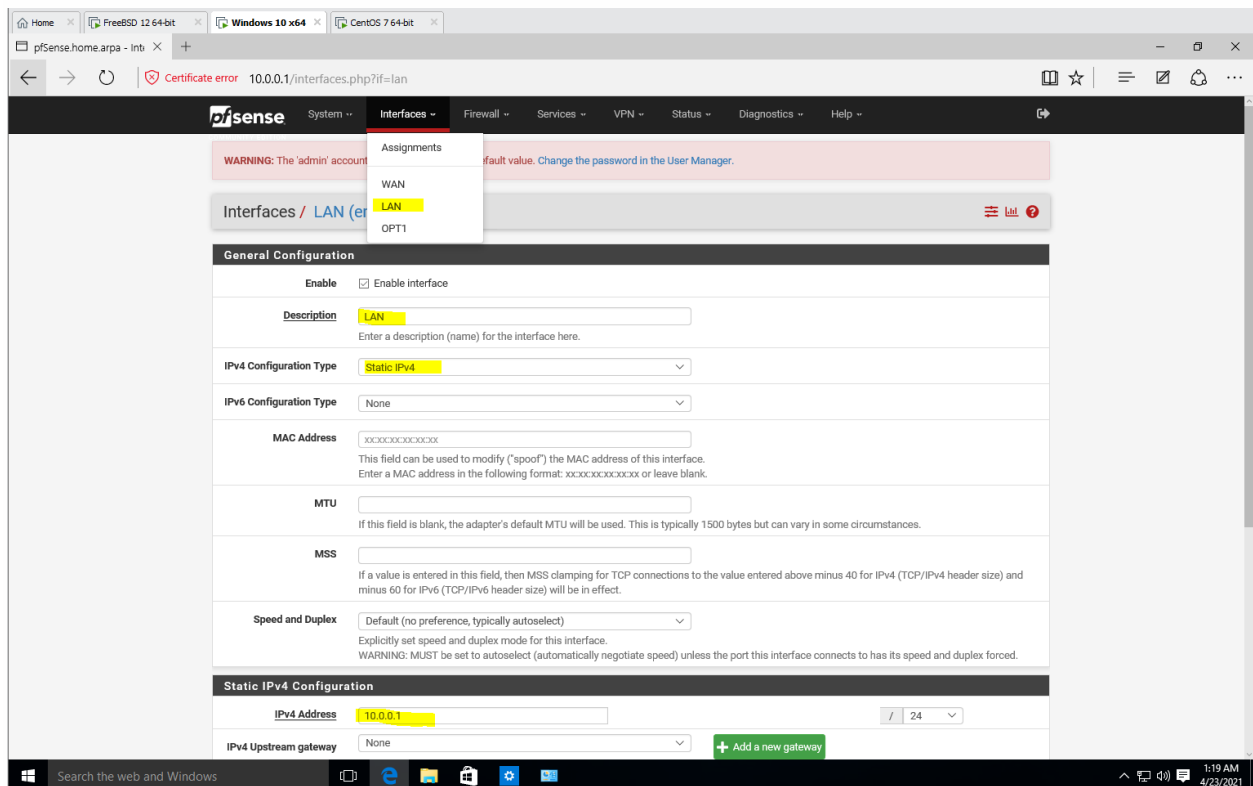


2. Cấu hình NAT

Interface WAN của pfsense



Interface LAN của pfSense



Cấu hình NAT cho phép các PC trong LAN có thể truy cập internet thông qua interface WAN

The screenshot shows the pfSense web interface for configuring Outbound NAT. The 'Firewall' menu is open, highlighting 'NAT'. The 'Outbound NAT Mode' section shows 'Automatic outbound NAT rule generation' selected. Below, the 'Mappings' table lists a rule for the WAN interface, source 10.0.0.0/24, and destination *. The 'Automatic Rules' section shows two rules: one for ISAKMP and another for general traffic. A green arrow points to the 'Add' button in the mappings section.

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Firewall / NAT / Outbound

Port Forward 1:1 Outbound NAT

Outbound NAT Mode

Mode

- ☒ Automatic outbound NAT rule generation. (IPsec passthrough included)
- ☐ Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)
- ☐ Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)
- ☐ Disable Outbound NAT rule generation. (No Outbound NAT rules)

[Save](#)

Mappings

	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
☒	WAN	10.0.0.0/24	*	*	*	WAN address	*			Add Delete Save

Automatic Rules:

	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description
☒	WAN	127.0.0.0/8 :: 1/128	10.0.0.0/24	20.0.0.0/24	*	WAN address	*	☒	Auto created rule for ISAKMP
☒	WAN	127.0.0.0/8 :: 1/128	10.0.0.0/24	20.0.0.0/24	*	WAN address	*		Auto created rule

Cho phép lớp 10.0.0.0/24 được NAT để đi ra internet

The screenshot shows the 'Edit Advanced Outbound NAT Entry' page. The 'Interface' is set to 'WAN'. The 'Address Family' is 'IPv4+IPv6'. The 'Protocol' is 'any'. The 'Source' is 'Network' with '10.0.0.0' and '24'. The 'Destination' is 'Any' with '24'. The 'Translation' section shows 'Address' set to 'Interface Address'.

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Firewall / NAT / Outbound / Edit

Edit Advanced Outbound NAT Entry

Disabled ☐ Disable this rule

Do not NAT ☐ Enabling this option will disable NAT for traffic matching this rule and stop processing Outbound NAT rules in most cases this option is not required.

Interface WAN
The interface on which traffic is matched as it exits the firewall. In most cases this is "WAN" or another externally-connected interface.

Address Family IPv4+IPv6
Select the Internet Protocol version this rule applies to.

Protocol any
Choose which protocol this rule should match. In most cases "any" is specified.

Source Network 10.0.0.0 / 24
Type Source network for the outbound NAT mapping. Port or Range

Destination Any / 24
Type Destination network for the outbound NAT mapping. Port or Range

☐ Not
Invert the sense of the destination match.

Translation

Address Interface Address
Connections matching this rule will be mapped to the specified Address. The Address can be an interface, a Host-type Alias, or a Virtual IP address.

Port or Range ☐ Static Port

nhấn “apply” để cấu hình

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

Firewall / NAT / Outbound

The NAT configuration has been changed.
The changes must be applied for them to take effect.

Port Forward 1:1 Outbound NAT

Outbound NAT Mode

Mode

- ☒ Automatic outbound NAT rule generation. (IPsec passthrough included)
- ☐ Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)
- ☐ Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)
- ☐ Disable Outbound NAT rule generation. (No Outbound NAT rules)

Save

Mappings

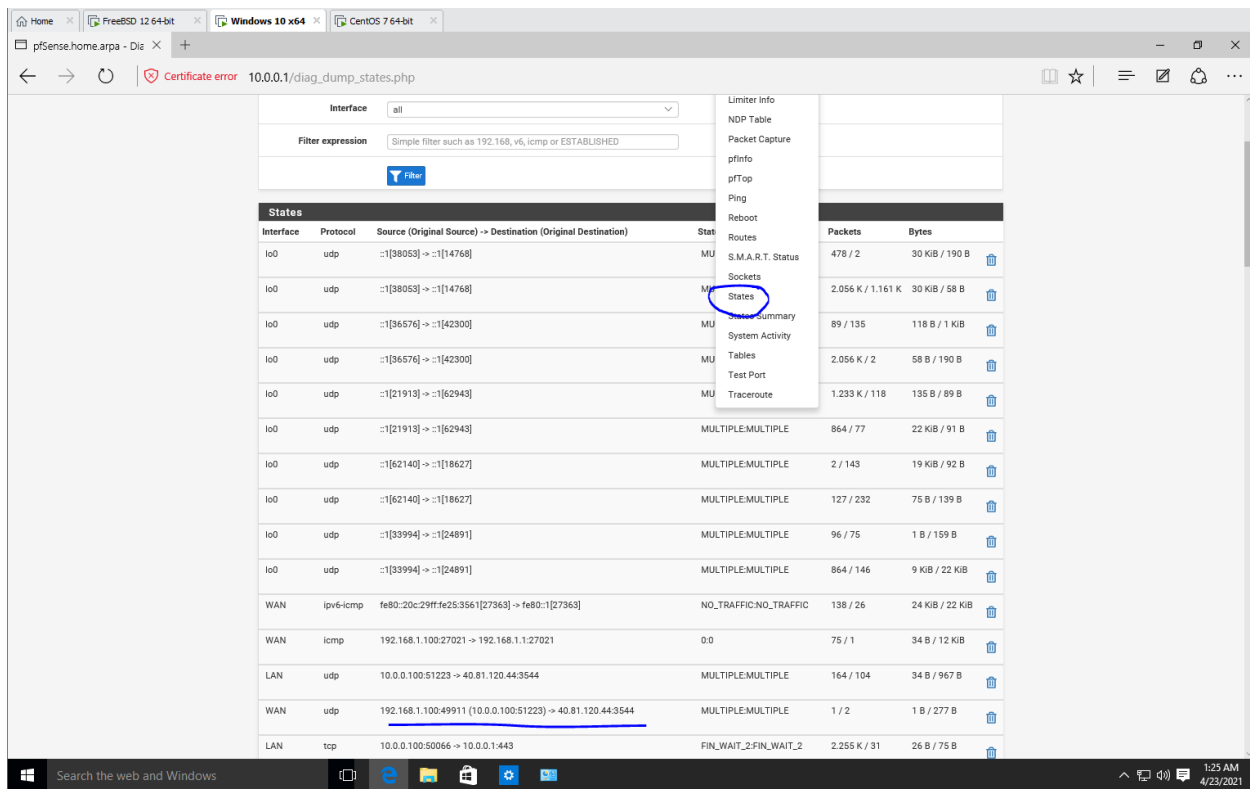
☐	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
☐	WAN	10.0.0.0/24	*	*	*	WAN address	*			Edit Delete Save

[Add](#) [Add](#) [Delete](#) [Save](#)

Automatic Rules:

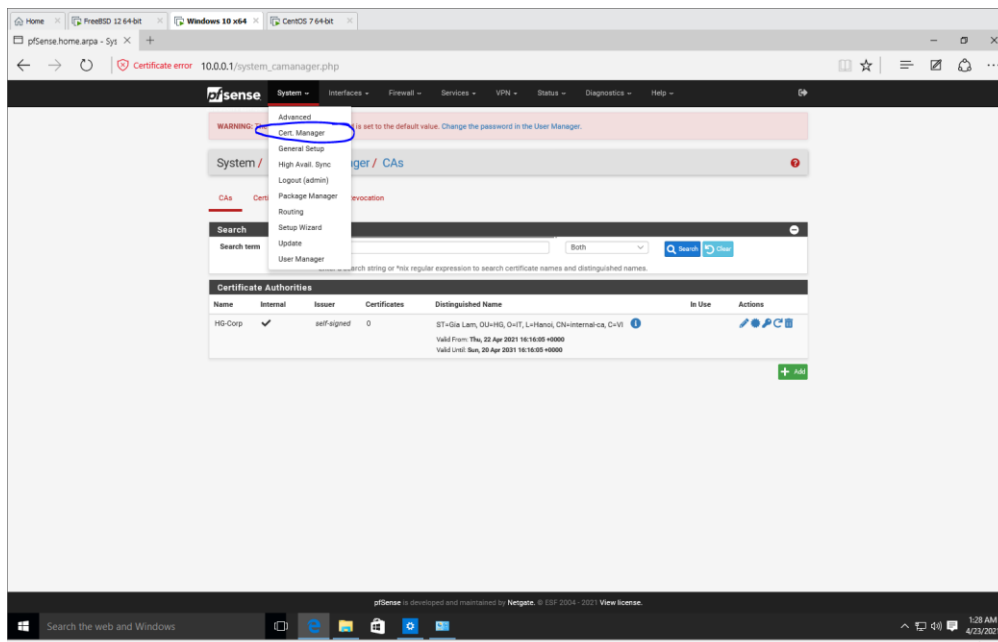
☐	Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description
☒	WAN	127.0.0.0/8 ::1/128	10.0.0.0/24	20.0.0.0/24	*	WAN address	*	☒	Auto created rule for ISAKMP
☒	WAN	127.0.0.0/8 ::1/128	10.0.0.0/24	20.0.0.0/24	*	WAN address	*		Auto created rule

có thể xem IP của các client được NAT tương ứng với port nào và đi đâu.

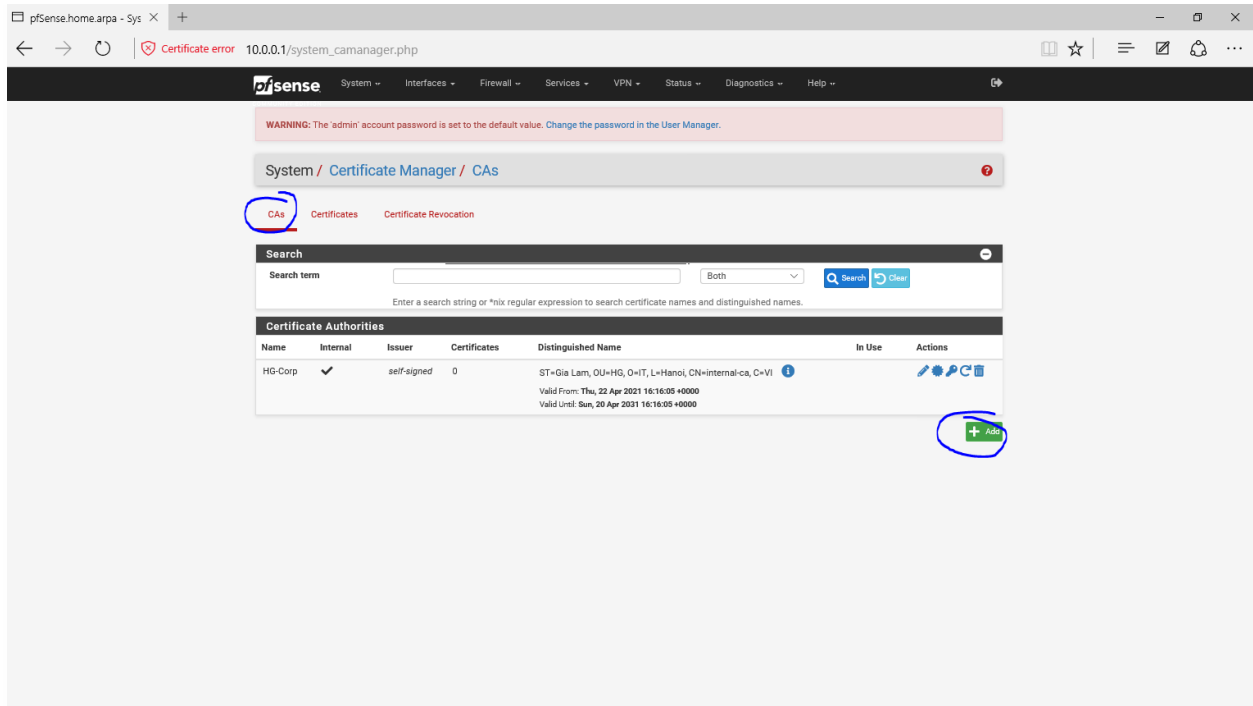


3. Cấu hình dịch vụ Squid Proxy Server trên firewall pfSense cắm máy Ubuntu truy cập Youtube.com hoặc Facebook.com, các website khác vẫn truy cập bình thường

Thao tác như sau: **System > Cert. Manager**



Chọn CAs > New



The screenshot shows the pfSense web interface at the URL `10.0.0.1/system_camanager.php`. The breadcrumb navigation is `System / Certificate Manager / CAs`. The `CAs` tab is selected and circled in blue. A warning message at the top states: "WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager." Below the navigation tabs, there is a search bar and a table titled "Certificate Authorities". The table has columns: Name, Internal, Issuer, Certificates, Distinguished Name, In Use, and Actions. One entry is listed: "HG-Corp" with a checked "Internal" box, "self-signed" issuer, 0 certificates, and a distinguished name "ST=Gia Lam, OU=HG, O=IT, L=Hanoi, CN=internal-ca, C=VI". The "In Use" column shows a blue circle with the number 1. The "Actions" column contains icons for edit, delete, and a green "+ Add" button, which is circled in blue.

Name	Internal	Issuer	Certificates	Distinguished Name	In Use	Actions
HG-Corp	☒	self-signed	0	ST=Gia Lam, OU=HG, O=IT, L=Hanoi, CN=internal-ca, C=VI Valid From: Thu, 22 Apr 2021 16:16:05 +0000 Valid Until: Sun, 20 Apr 2031 16:16:05 +0000	1	

Điền tên cho CA và chọn **Method: Create an internal Certificate Authority**, sau đó điền đầy đủ thông tin của bạn vào các trường.

Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

pfSense.home.arpa - Sys x +

10.0.0.1/system_canager.php?act=new

Create / Edit CA

Descriptive name Test

Method Create an internal Certificate Authority

Trust Store ☐ Add this Certificate Authority to the Operating System Trust Store
When enabled, the contents of the CA will be added to the trust store so that they will be trusted by the operating system.

Randomize Serial ☐ Use random serial numbers when signing certificates
When enabled, if this CA is capable of signing certificates then serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using the sequential value from Next Certificate Serial.

Internal Certificate Authority

Key type RSA

2048
The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm sha256
The digest method used when the CA is signed.
The best practice is to use an algorithm stronger than SHA1. Some platforms may consider weaker digest algorithms invalid

Lifetime (days) 3650

Common Name internal-ca

The following certificate authority subject components are optional and may be left blank.

Country Code VI

State or Province e.g. Texas

City HCM

Organization IT

Organizational Unit Test

Save

Search the web and Windows

1:34 AM
4/23/2021

Thông tin của CA mới được hiển thị.

Home x FreeBSD 12 64-bit x Windows 10 x64 x CentOS 7 64-bit x

pfSense.home.arpa - Sys x +

10.0.0.1/system_canager.php

pfSense System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

System / Certificate Manager / CAs

CA's Certificates Certificate Revocation

Search

Search term Both Search Clear

Enter a search string or *nix regular expression to search certificate names and distinguished names.

Certificate Authorities

Name	Internal	Issuer	Certificates	Distinguished Name	In Use	Actions
HG-Corp	✓	self-signed	0	ST=Gia Lam, OU=HG, O=IT, L=Hanoi, CN=internal-ca, C=VI Valid From: Thu, 22 Apr 2021 16:16:05 +0000 Valid Until: Sun, 20 Apr 2021 16:16:05 +0000	1	
Test	✓	self-signed	0	OU=Test, O=IT, L=HCM, CN=internal-ca, C=VI Valid From: Thu, 22 Apr 2021 18:35:07 +0000 Valid Until: Sun, 20 Apr 2021 18:35:07 +0000	1	

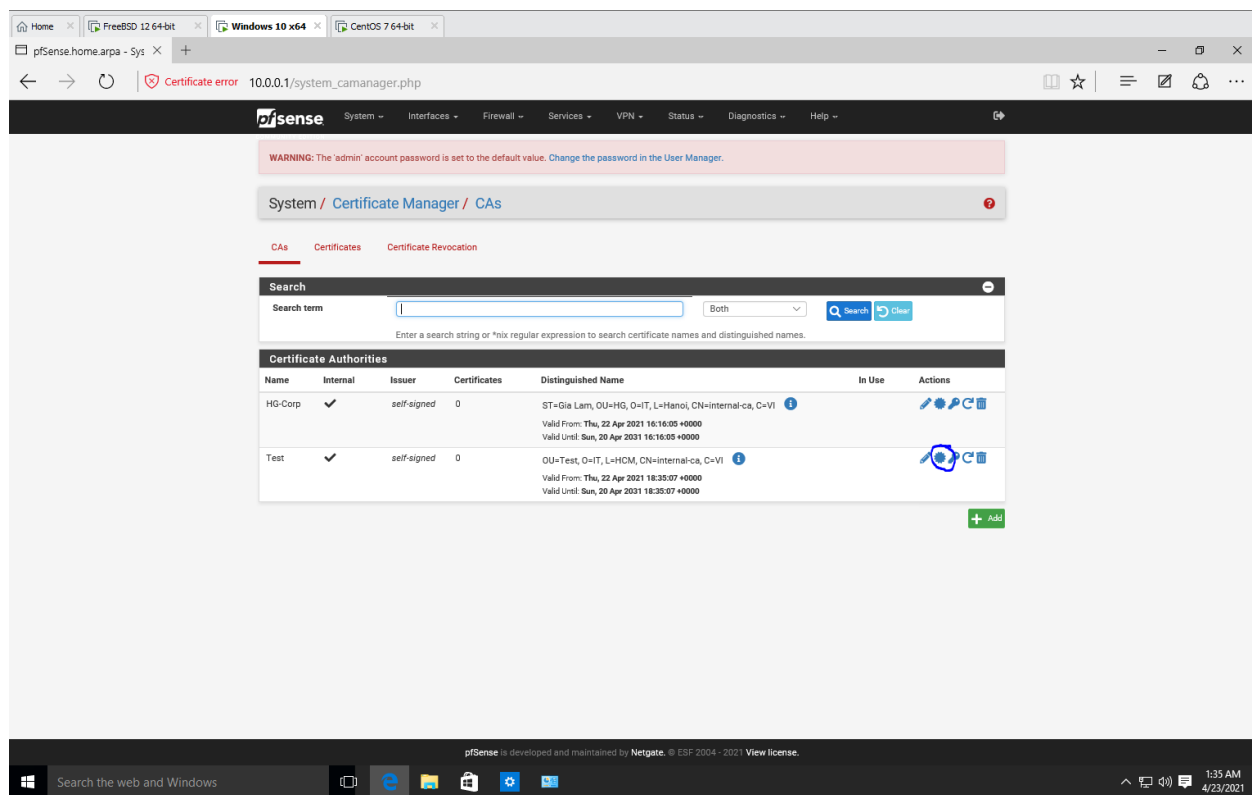
+ Add

pfSense is developed and maintained by Netgate. © ESF 2004 - 2021 View license.

Search the web and Windows

1:35 AM
4/23/2021

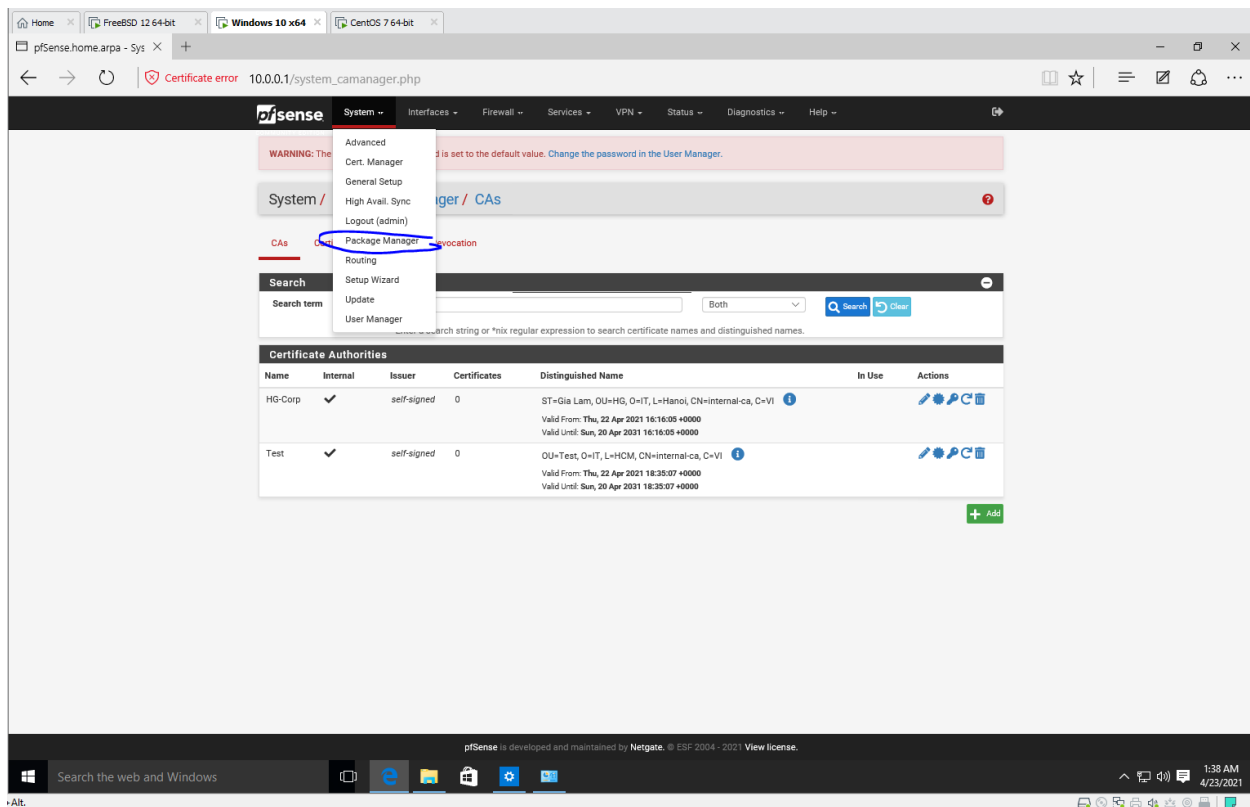
Nhấp vào biểu tượng hình sao (**Export CA**) để tải CA về và copy vào các máy client.



Cài đặt gói Squid và SquidGuard

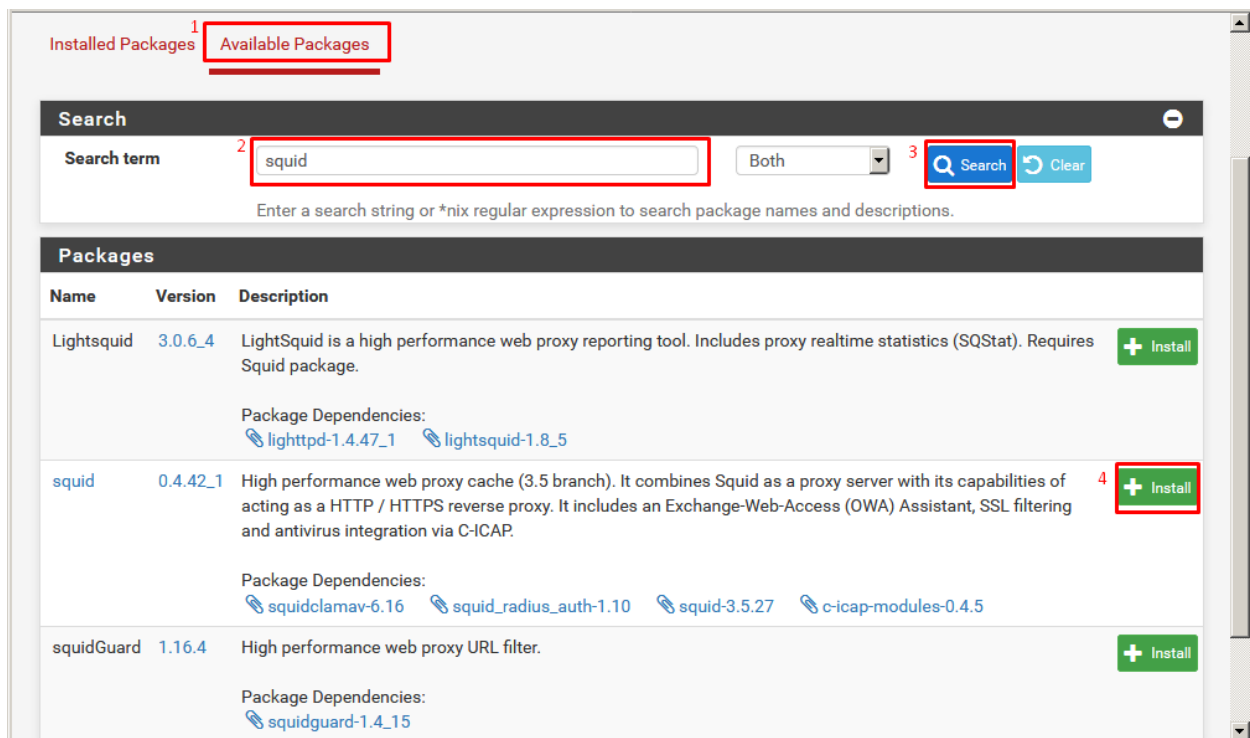
cài đặt gói squid - một Proxy Server khá nổi tiếng. Thao tác như sau:

vào **System > Package Manager**



Tìm kiếm gói cài đặt

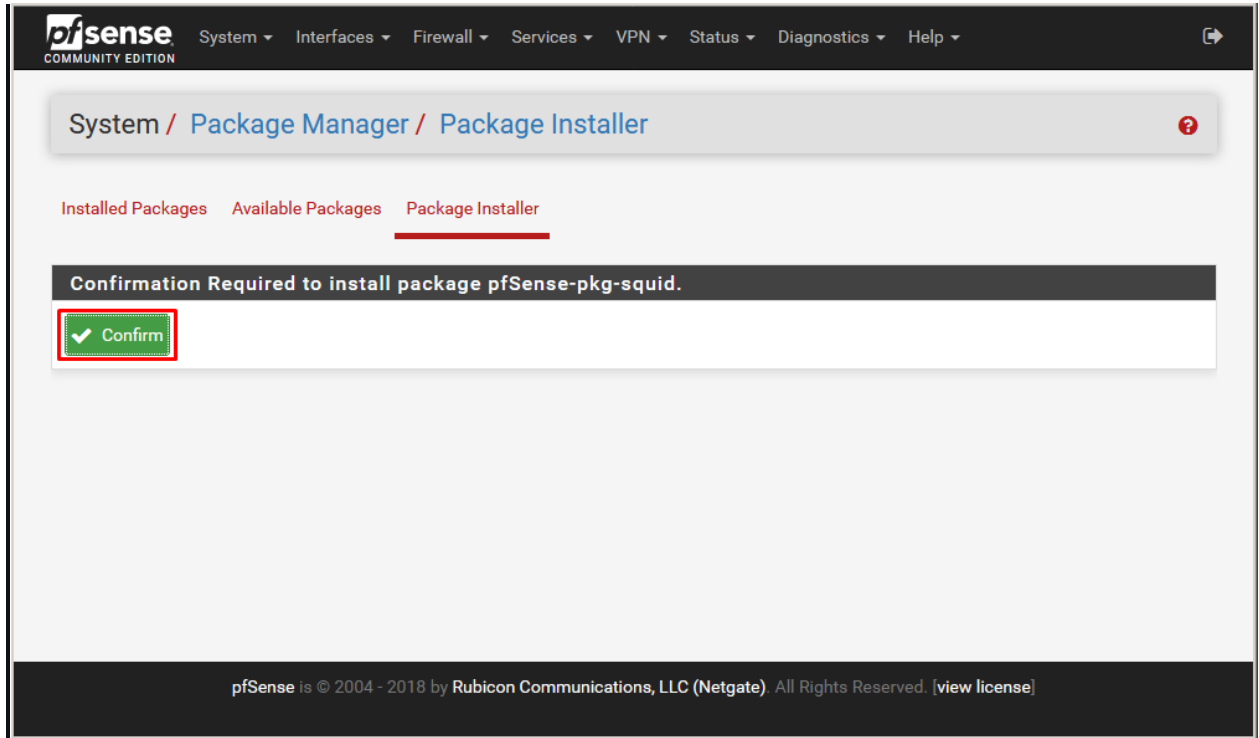
Chọn tab Available Package, gõ squid vào ô tìm kiếm và bấm Search



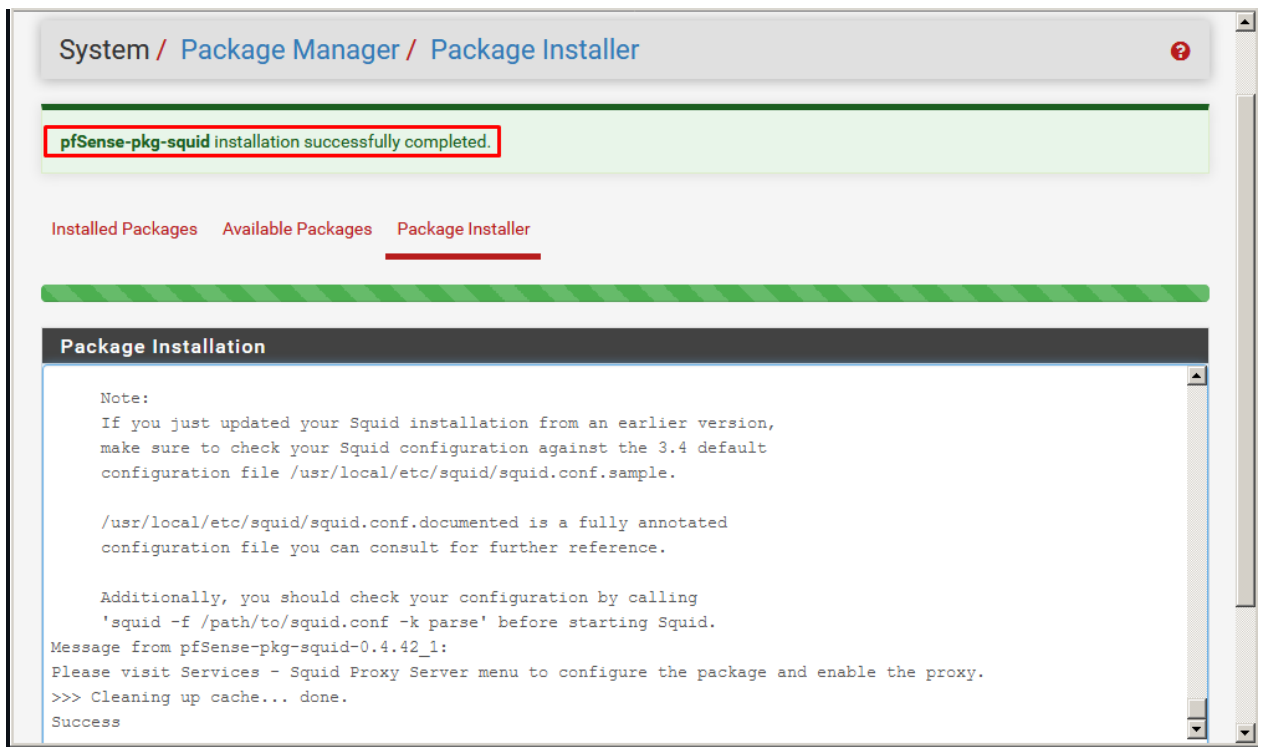
Cài đặt 2 gói squid và squidGuard

cài đặt lần lượt 2 gói squid và squidGuard bằng cách bấm vào nút + Install.

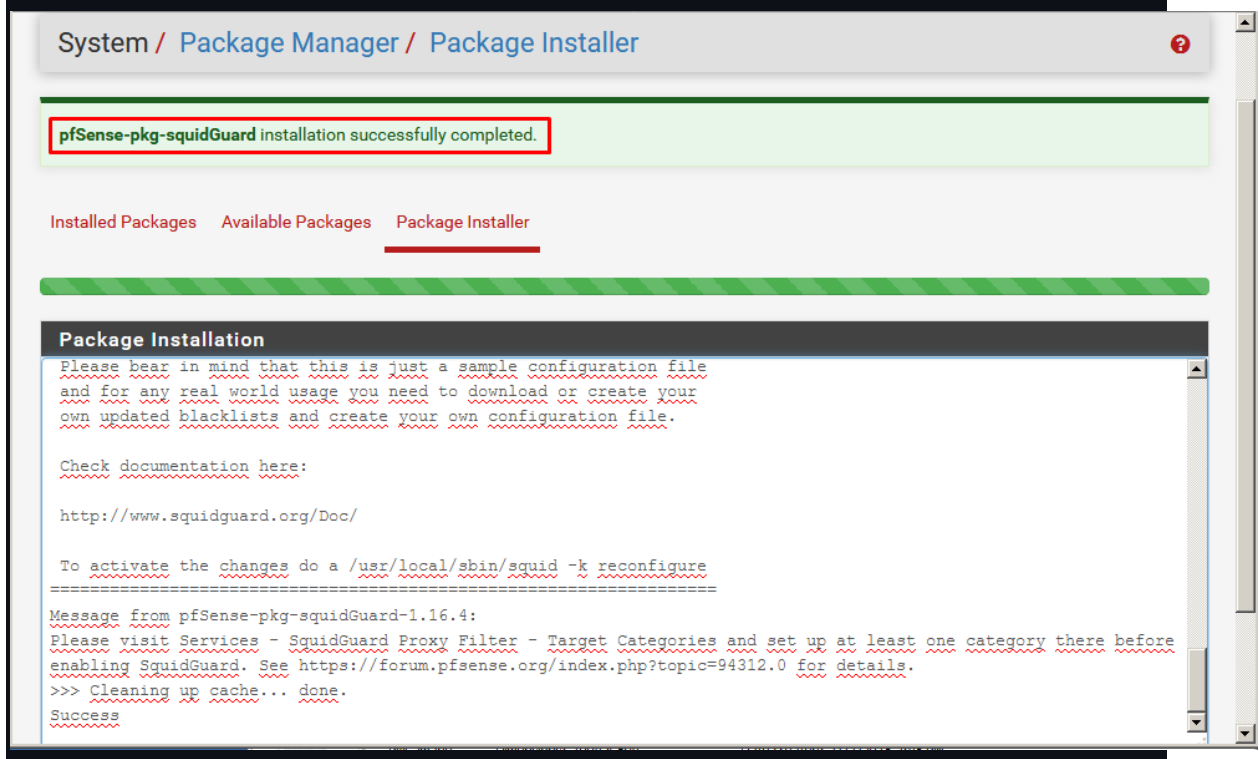
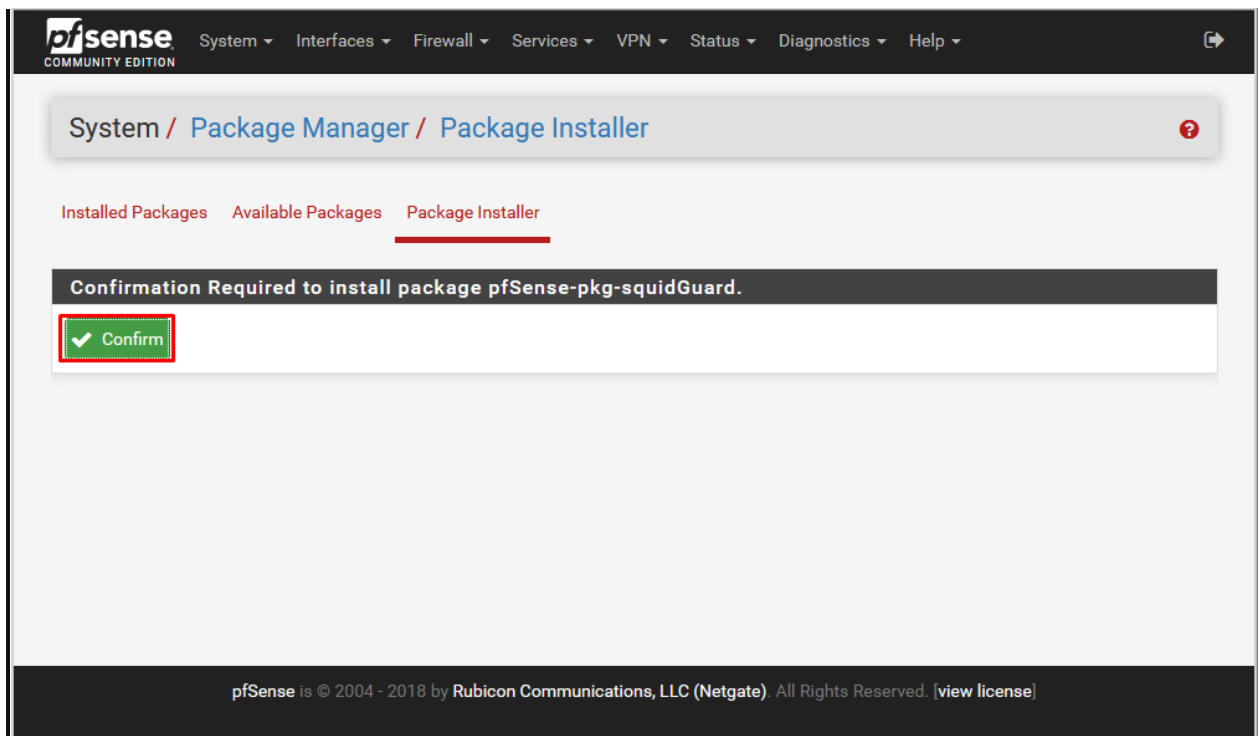
Bấm **Confirm** để xác nhận cho việc cài đặt.



Chờ vài phút, squid đã cài xong.



tìm kiếm và xác nhận cài đặt gói squidGuard.



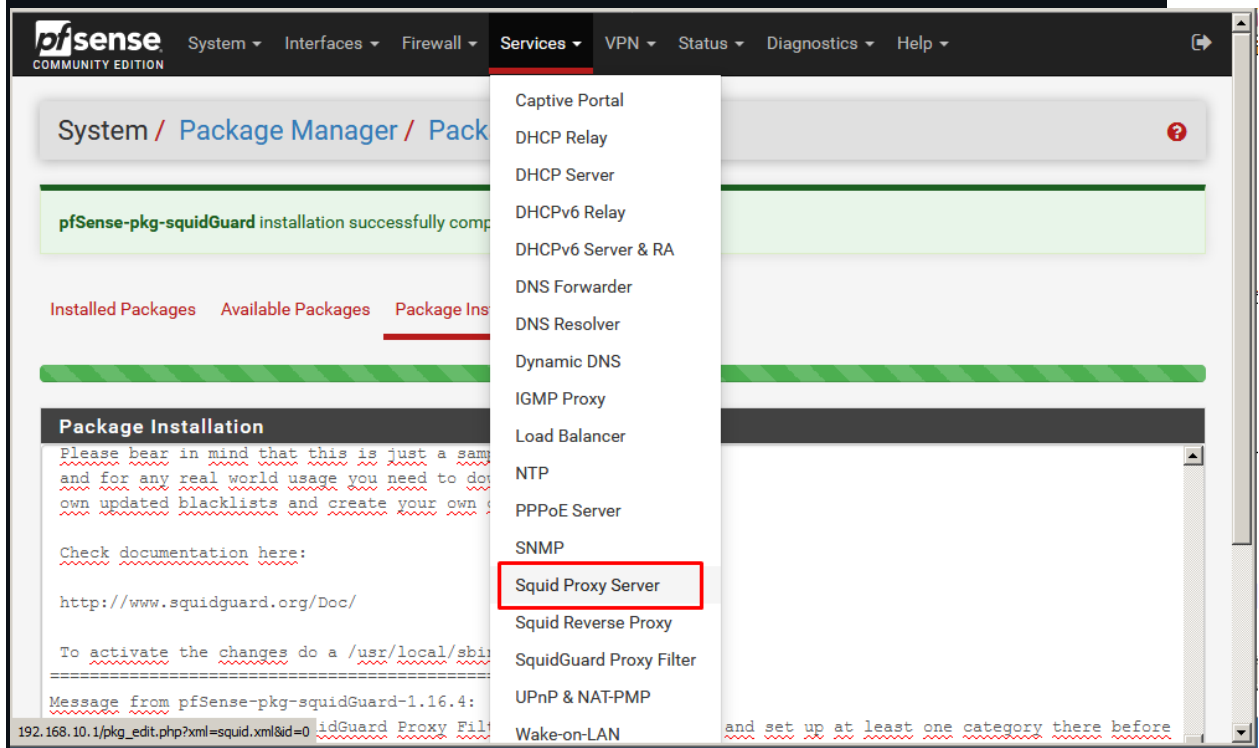
chuyển sang bước tiếp theo để cấu hình squid.

Cấu hình squid

cấu hình cơ bản cho squid - Proxy Server.

Mở phần cấu hình của squid

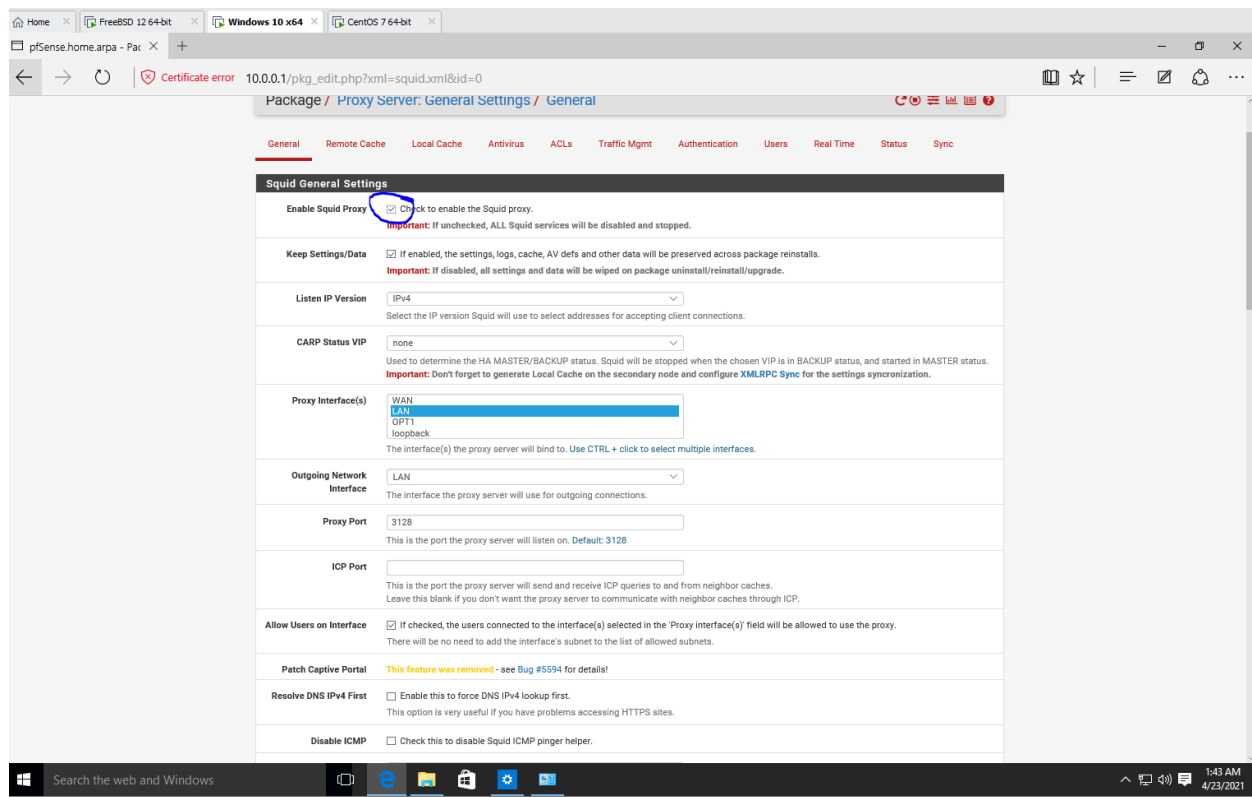
Trên Menu, chúng ta chọn **Services > Squid Proxy Server**



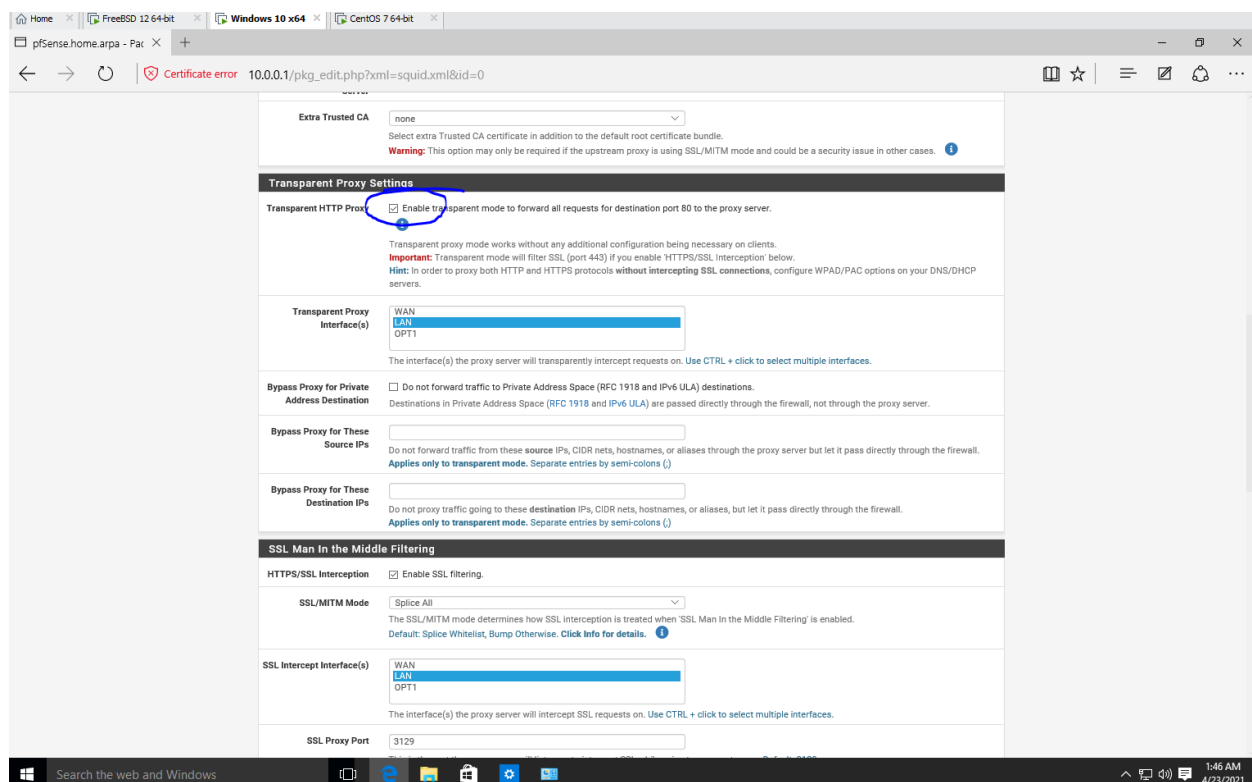
Cấu hình squid

Kích hoạt squid

Chọn tab General, tích vào ô "Check to enable the Squid proxy."



Cấu hình "Transparent Proxy Settings", tùy chọn này cho phép tất cả các luồng traffic từ client sẽ qua cổng 80 thay vì 3218 mặc định. Vì vậy, không phải khai báo cấu hình trên Client.



Bật tính năng **SSL Man In the Middle Filtering** và chọn CA đã tạo và không kiểm tra Certificate từ phía client.

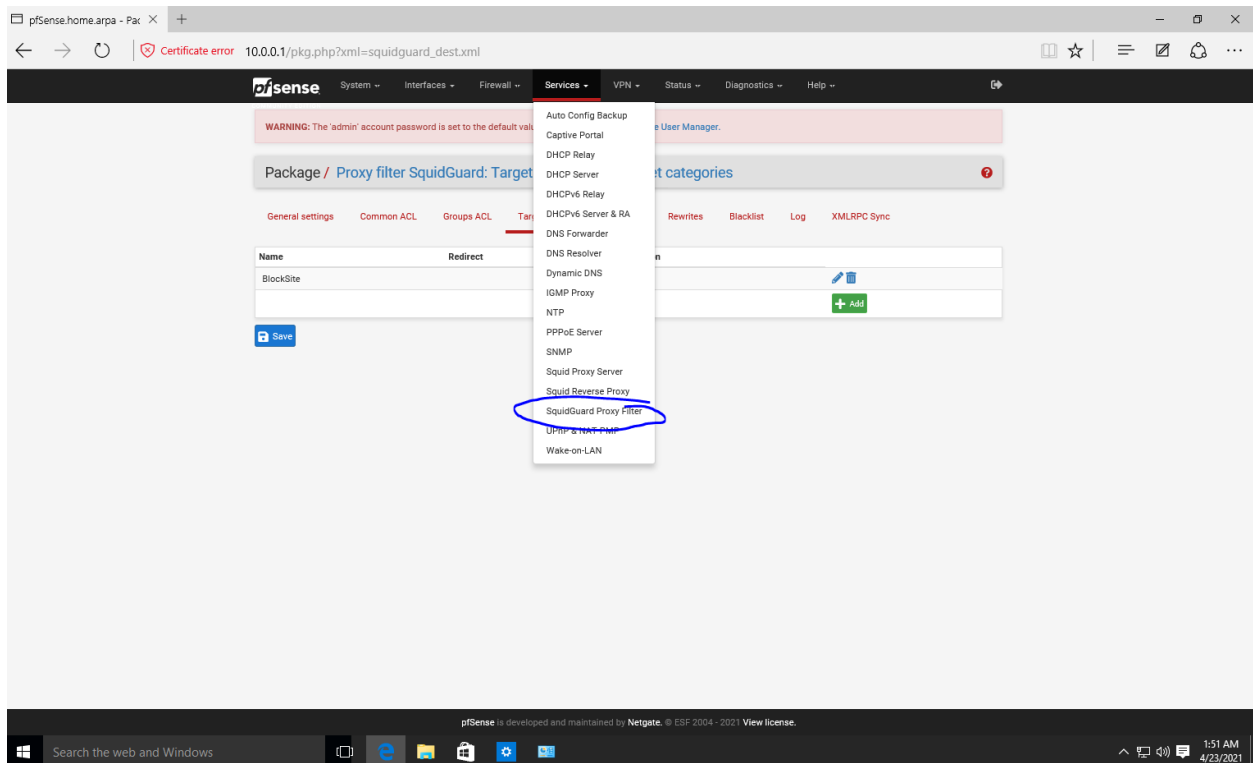
The screenshot shows the 'SSL Man In the Middle Filtering' configuration page in pfSense. The 'Bypass Proxy for Private Address Destination' section is expanded, showing options to bypass proxy for source and destination IPs. The 'SSL Man In the Middle Filtering' section is active, with 'Enable SSL filtering' checked. The 'SSL/MITM Mode' is set to 'Splice All'. The 'SSL Intercept Interface(s)' are set to 'WAN', 'LAN', and 'OPT1'. The 'SSL Proxy Port' is 3129. The 'SSL Proxy Compatibility Mode' is 'Modern'. The 'DHParams Key Size' is '2048 (default)'. The 'CA' is set to 'Test'. The 'SSL Certificate Daemon Children' is 5. The 'Remote Cert Checks' are set to 'Accept remote server certificate with errors' and 'Do not verify remote certificate'. The 'Certificate Adapt' options are 'Sets the "Not After" (setValidAfter)' and 'Sets the "Not Before" (setValidBefore)'. The 'Certificate Adapt' section is expanded, showing the 'Set the "Not After" (setValidAfter)' and 'Set the "Not Before" (setValidBefore)' options.

Bật tính năng ghi Log và cấu hình log xoay vòng trong 7 ngày và lưu lại

The screenshot shows the 'Logging Settings' and 'Headers Handling, Language and Other Customizations' configuration pages in pfSense. The 'Logging Settings' section is active, with 'Enable Access Logging' checked. The 'Log Store Directory' is '/var/squid/logs'. The 'Rotate Logs' is set to 7. The 'Log Pages Denied by SquidGuard' is unchecked. The 'Headers Handling, Language and Other Customizations' section is active, with 'Visible Hostname' set to 'localhost', 'Administrator's Email' set to 'admin@localhost', 'Error Language' set to 'en', 'X-Forwarded Header Mode' set to '(on)', 'Disable VIA Header' unchecked, 'URI Whitespace Characters Handling' set to 'strip', and 'Suppress Squid Version' unchecked. The 'Save' button is highlighted.

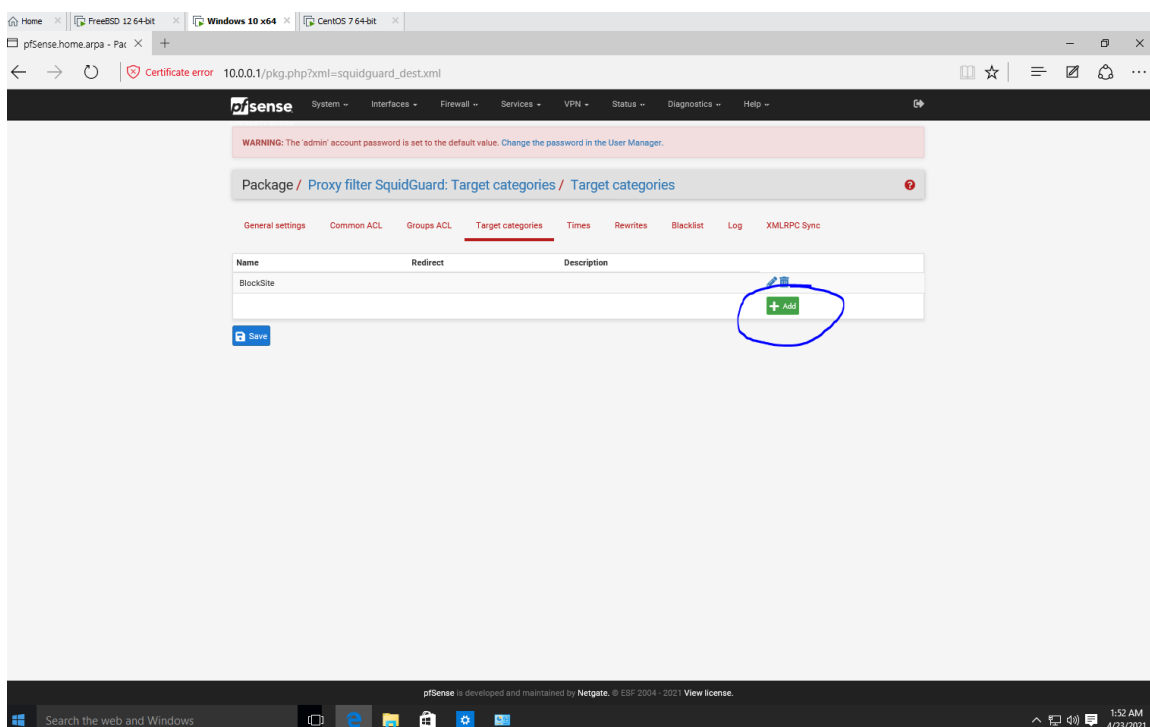
Cấu hình squidGuard

Sau khi hoàn thành phần cấu hình squid, chúng ta sẽ cấu hình squidGuard để xác định những danh sách những địa chỉ nào được phép, địa chỉ nào bị cấm truy cập.



Tạo danh sách các website bị cấm truy cập

Chọn tab Target categories > Add



Đặt tên cho Category và liệt kê các domain bị chặn. Trong ví dụ này, chúng tôi chặn 2 domain là facebook.com và youtube.com.

The screenshot shows the pfSense web interface for configuring SquidGuard target categories. At the top, there is a warning message: "WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager." Below this, the breadcrumb navigation reads "Proxy filter SquidGuard: Target categories / Edit / Target categories". The main navigation bar includes links for "General settings", "Common ACL", "Groups ACL", "Target categories" (which is highlighted), "Times", "Rewrites", "Blacklist", "Log", and "XMLRPC Sync". The "General Options" section contains several fields: "Name" (set to "BlockSite"), "Order" (a dropdown menu), and "Domain List" (containing "facebook.com youtube.com"). Below the "Domain List" is a text box for "URL List".

bên dưới tích vào tùy chọn ghi lại Log và lưu.

This screenshot shows the bottom portion of the pfSense SquidGuard configuration page. It includes fields for "Regular Expression" and "Redirect mode" (set to "none"). Below these is a "Redirect" field and a "Description" field. At the bottom, there is a "Log" section with a checkbox labeled "Check this option to enable logging for this ACL", which is checked. A blue circle highlights the "Save" button at the bottom of the form. The footer of the page indicates "pfSense is developed and maintained by Netgate. © ESF 2004 - 2021 View license." and shows the system time as 1:54 AM on 4/23/2021.

Áp dụng Deny cho category

Chọn tab Common ACL và áp dụng chính sách Deny

General Options

Target Rules: !BlockSite all

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories	access
[BlockSite]	deny
Default access [all]	allow

Do not allow IP-Addresses in URL: ☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Proxy Denied Error:

Redirect mode: int error page (enter error message)
 Select redirect mode here.
 Note: if you use 'transparent proxy', then 'int' redirect mode will not be accessible.
 Options: [ext url err page](#), [ext url redirect](#), [ext url as 'move'](#), [ext url as 'found'](#)

Redirect info:

Use SafeSearch engine: ☐ Enable the protected mode of search engines to limit access to mature content.
 At the moment it is supported by Google, Yandex, Yahoo, MSN, Live Search, Bing, DuckDuckGo, OneSearch, Rambler, Ecosia and Qwant. Make sure that the search engines can be accessed. It is recommended to prohibit access to others.
 Note: This option overrides 'Rewrite' setting.

Bật tùy chọn ghi log và lưu lại cấu hình.

General Options

Target Rules: !BlockSite all

Target Rules List + -

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

Target Categories	access
[BlockSite]	deny
Default access [all]	allow

Do not allow IP-Addresses in URL: ☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

Proxy Denied Error:

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 Note: This option overrides 'Rewrite' setting.

Rewrite: none (rewrite not defined)
 Enter the rewrite condition name for this rule or leave it blank.

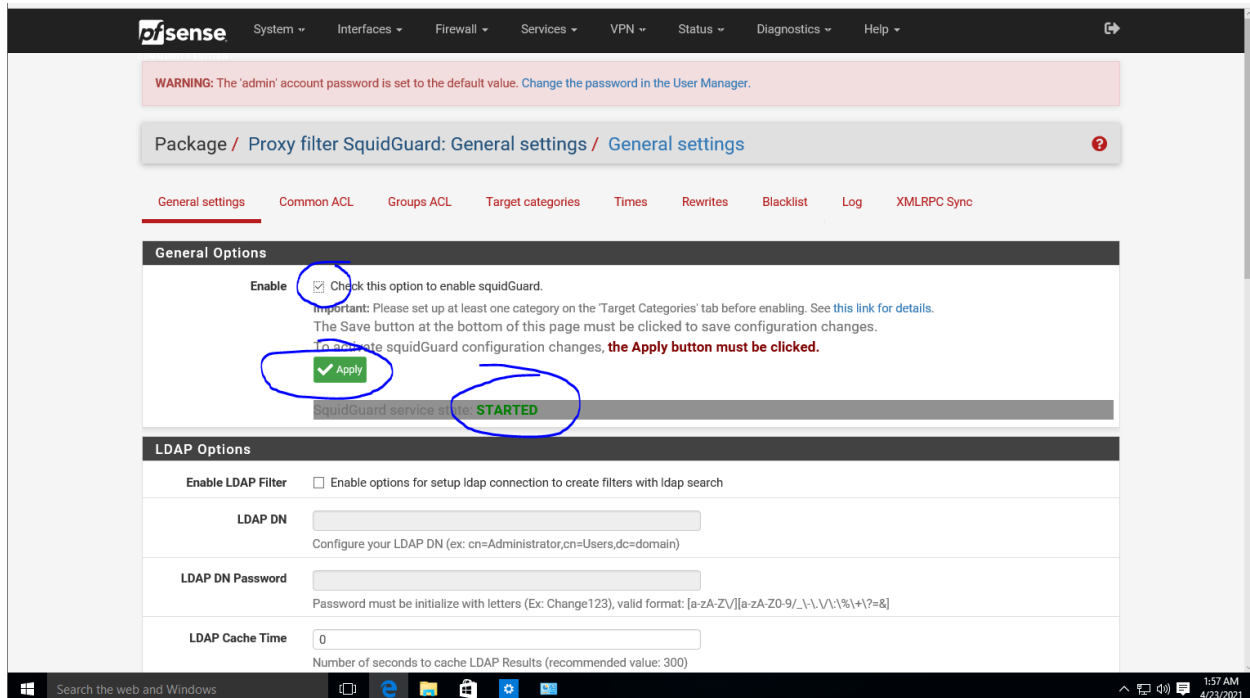
Log: ☒ Check this option to enable logging for this ACL.

Save

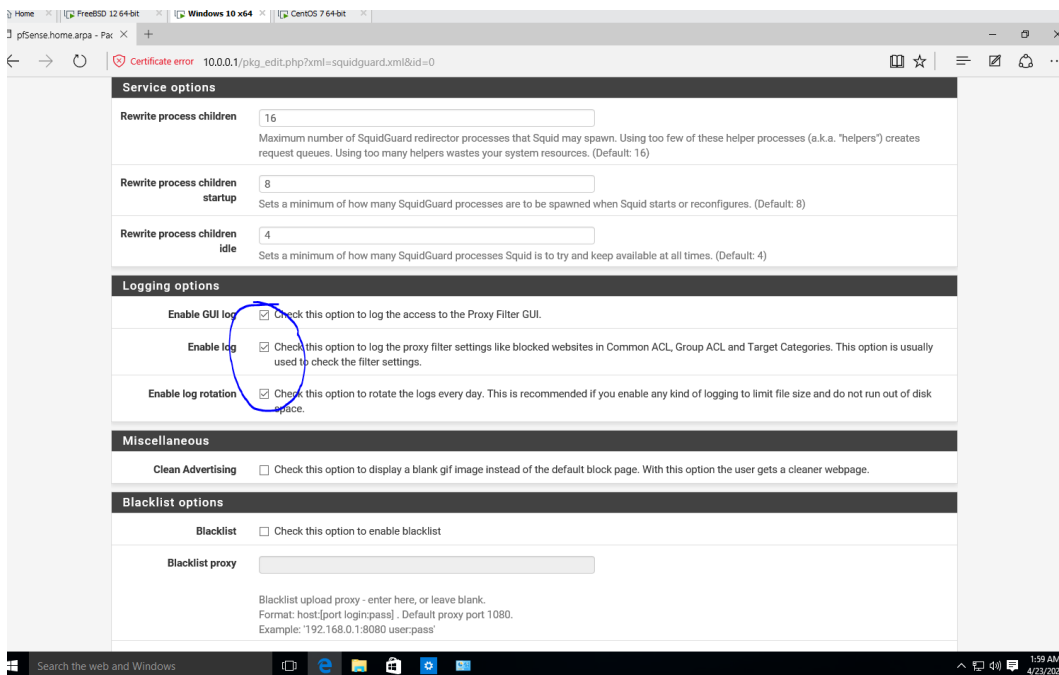
Kích hoạt squidGuard

Tích vào tùy chọn và bấm vào Apply.

Lưu ý: Mỗi khi có thay đổi ở phần Target categories, chúng ta phải bấm vào nút Apply để áp dụng những thay đổi.

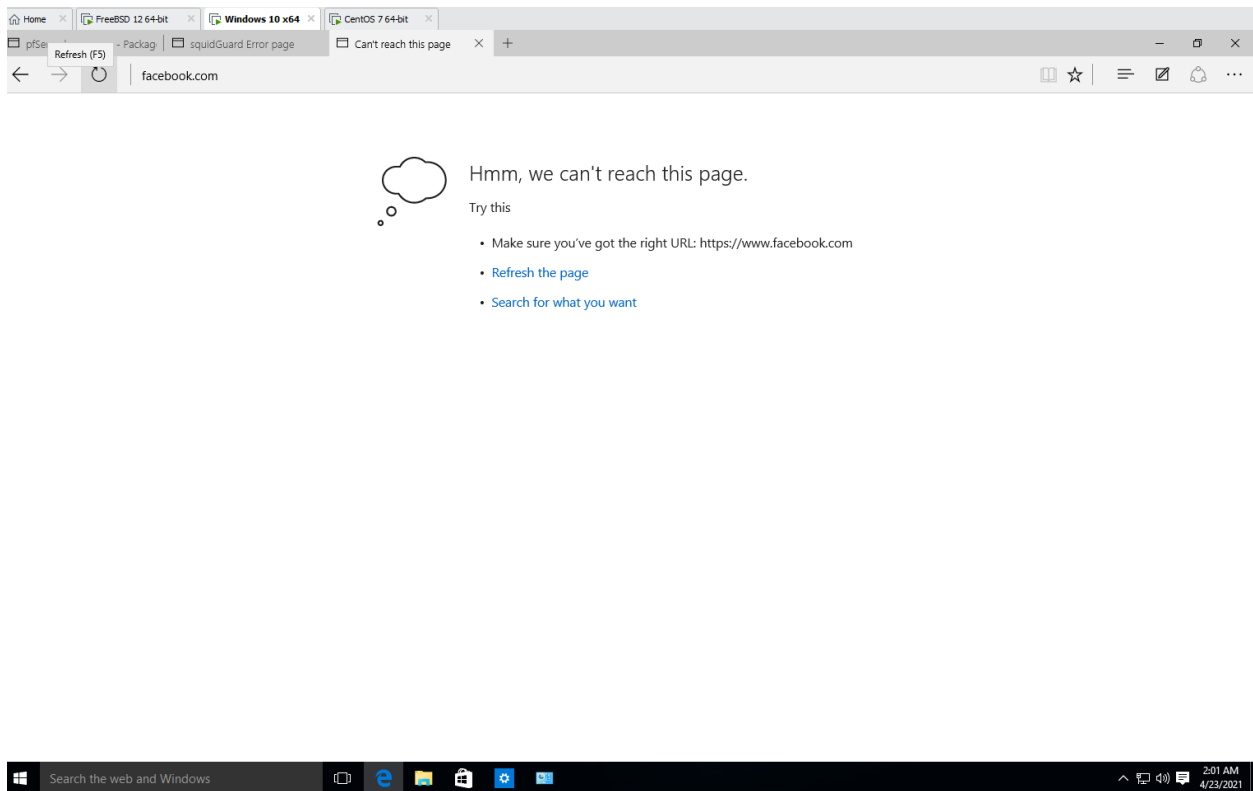


Cấu hình thêm về squidGuard ghi log và lưu lại cấu hình

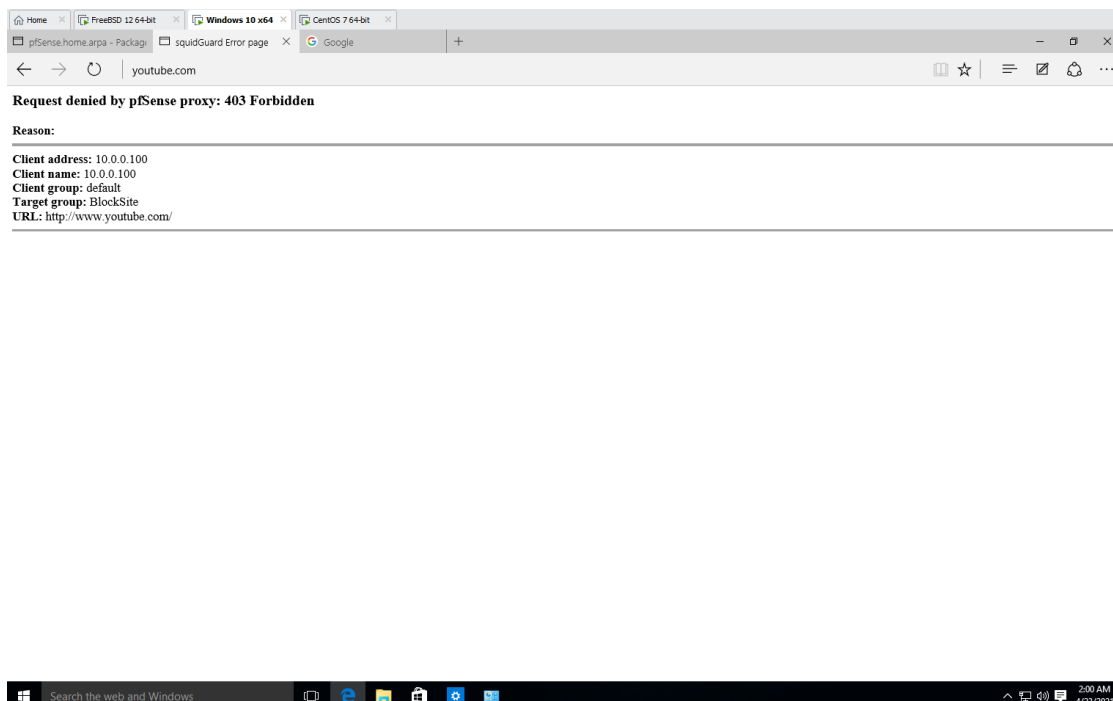


kiểm tra

Khi truy cập vào facebook.com



Khi truy cập vào youtube.com



Khi truy cập vào google.com

